

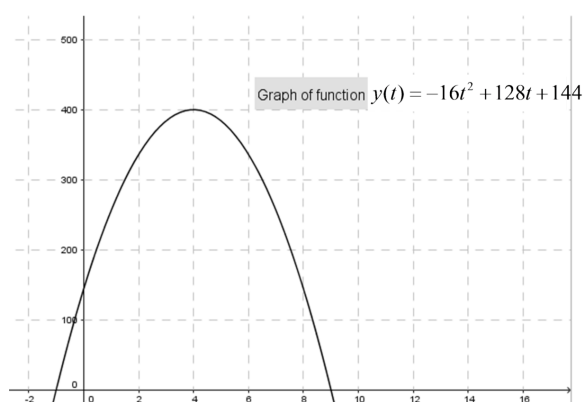
Problem 2

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Problem 1 Part of the given parabola can be used to model the “position-time” graph of a ball thrown straight up into the air. The graph gives the height of the ball in feet t seconds after being thrown into the air.

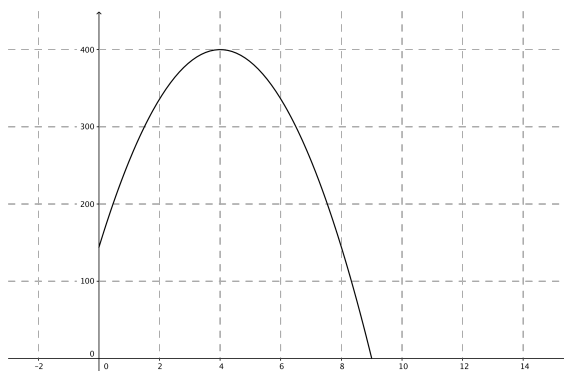
Let the function f be defined by $f(t) = -16t^2 + 128t + 144$.

Use this graph, and the given function, f , to answer the following questions.



- (a) Mark the part of the parabola that can be used to model the position of the ball.

Explanation.



- (b) What are the units on the t axis? What are the units on the y axis?

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Explanation. The units on the t axis are “seconds” (for time), while the units on the y axis are “feet” (for height).

- (c) *If you were watching a movie of the ball being thrown, is the graph a picture of the path that the ball follows? Why or why not?*

Explanation. No, the position-time graph is *not* the path the ball follows. The graph shows the height of the ball at a given *time*. The ball is thrown straight up and has no horizontal movement so its path is on a vertical line.

- (d) *Let $f(t)$ denote the height of the ball at time $t, t \geq 0$. What is the height of the ball at time $t = 0$?*

Explanation. The height can be found by finding $f(0)$

$$f(0) = -16(0)^2 + 128(0) + 144$$

$$f(0) = 144 \text{ feet}$$

- (e) *When will the ball hit the ground?*

Explanation. The ball will hit the ground when the height $f(t)$ equals zero.

$$0 = -16t^2 + 128t + 144$$

$$0 = -16(t^2 - 8t - 9)$$

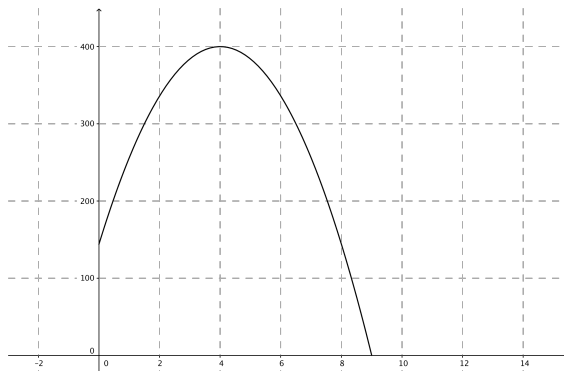
$$0 = -16(t + 1)(t - 9)$$

$$t = -1, 9$$

The ball will hit the ground at $t = 9$ or 9 seconds after the ball is thrown into the air.

- (f) *What is the domain of the position function, f , of the ball?*

Explanation. The domain of f is the interval $[0, 9]$. The ball is thrown straight up at $t = 0$ and hits the ground at $t = 9$. With this domain, the position-time graph of the ball is given by



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- (g) Use the table of values to find the average velocity of the ball between $t = 8.9$ and $t = 9$ seconds.

t	$\approx f(t)$
8.9	15.84
8.99	1.6
8.999	0.159984
8.9999	0.015998
9	0

Explanation. The average velocity of the ball between $t = 8.9$ seconds and $t = 9$ seconds is

$$\frac{f(9) - f(8.9)}{9 - 8.9} = \frac{0 - 15.84}{0.1} = -158.4 \text{ feet per second}$$

- (h) Use the table of average velocities to approximate the instantaneous velocity of the ball when it hits the ground.

Time Interval	Average Velocity
$[8.9, 9]$	$\frac{f(9) - f(8.9)}{.1} = \frac{0 - 15.84}{.1} = -158.4$
$[8.99, 9]$	$\frac{f(9) - f(8.99)}{.01} = \frac{0 - 1.5984}{.01} = -159.84$
$[8.999, 9]$	$\frac{f(9) - f(8.999)}{.001} = \frac{0 - .159984}{.001} = -159.984$
$[8.9999, 9]$	$\frac{f(9) - f(8.9999)}{.0001} = \frac{0 - .0159998}{.0001} = -159.998$

Explanation. The instantaneous velocity of the ball hitting the ground appears to be -160 ft/sec.

- (i) Compute $v_{AV(t)}$ the average velocity of the ball on the time interval $[t, 9]$, where $t < 9$.

Explanation.

$$\begin{aligned}
 v_{AV(t)} &= \frac{f(9) - f(t)}{9 - t} \\
 &= \frac{0 - (-16t^2 + 128t + 144)}{9 - t} \\
 &= \frac{16t^2 - 128t - 144}{9 - t} \\
 &= \frac{16(t^2 - 8t - 9)}{9 - t} \\
 &= \frac{16(t - 9)(t + 1)}{9 - t} \\
 &= \frac{-16(9 - t)(t + 1)}{9 - t} \\
 &= -16(t + 1) \text{ feet per second}
 \end{aligned}$$

(j) Compute $v(9)$, the **instantaneous** velocity of the ball at $t = 9$.

Explanation.

$$\begin{aligned}
 v(9) &= \lim_{t \rightarrow 9^-} v_{AV(t)} \\
 &= \lim_{t \rightarrow 9^-} \frac{f(9) - f(t)}{9 - t}
 \end{aligned}$$

Notice that this is an indeterminate form, with form $\frac{0}{0}$.

$$\begin{aligned}
 v(9) &= \lim_{t \rightarrow 9^-} v_{AV(t)} \\
 &= \lim_{t \rightarrow 9^-} \frac{f(9) - f(t)}{9 - t} \\
 &= \lim_{t \rightarrow 9^-} -16(t + 1) \\
 &= -16(9 + 1) \\
 &= -160 \text{ feet per second}
 \end{aligned}$$